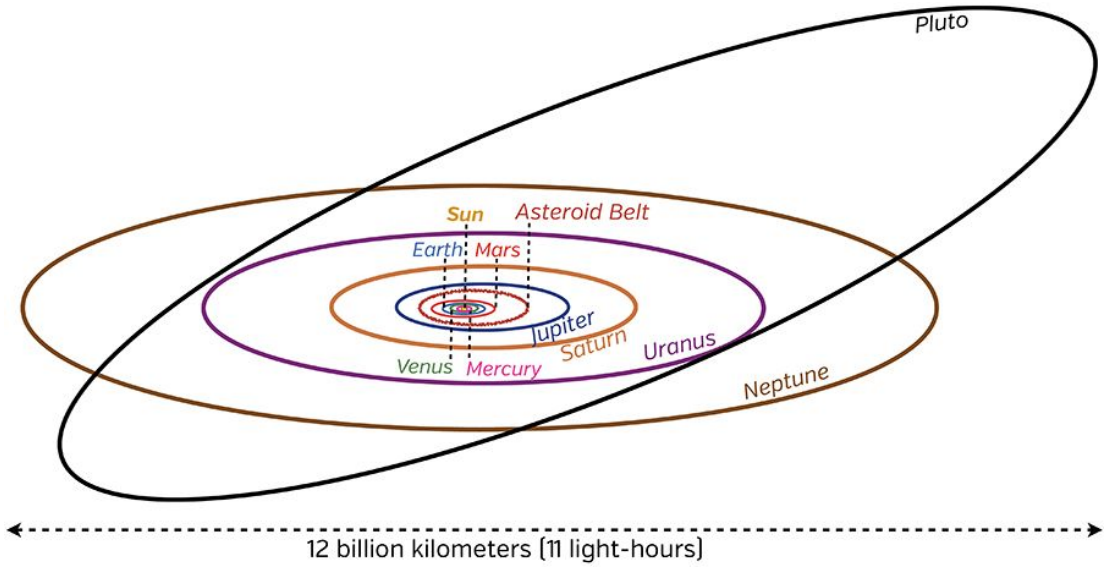


Fig. 2.4. The orbits of the Sun's planets and Pluto, and a region containing many asteroids.

The solar system is a thousand times bigger than the Sun itself; light needs eleven hours to travel across it.

The distance to the nearest star other than the Sun, Proxima Centauri, is 4.24 light-years, 2500 times farther than the distance across the solar system! In Chapter 13, I discuss the awful implications for interstellar travel.

Stellar Death: White Dwarfs, Neutron Stars, and Black Holes

The Sun and Earth are about 4.5 billion years old, about a third the age of the universe. After another 6.5 billion years or so, the Sun will exhaust the nuclear fuel in its core, the fuel that keeps it hot. The Sun then will shift to burning fuel in a shell around its core, and its surface will expand to engulf and fry the Earth. With the shell's fuel spent and the Earth fried, the Sun will shrink to become a white dwarf star, about the size of the Earth but with density a million times higher. The white